

Docker Task1

Provisioned new EC2 instance for docker practice

Instances (1/1) Info

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

Connect

Instance state

Actions

Launch instances

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
☒	docker-ec2	i-0e82aae401e6e51a0	Running	t2.micro	2/2 checks pass	View alarms	us-east-1d	ec2-13-221-139-24

Docker installed

```
ubuntu@ip-172-31-0-178:~$ sudo apt install docker.io
Installing:
  docker.io

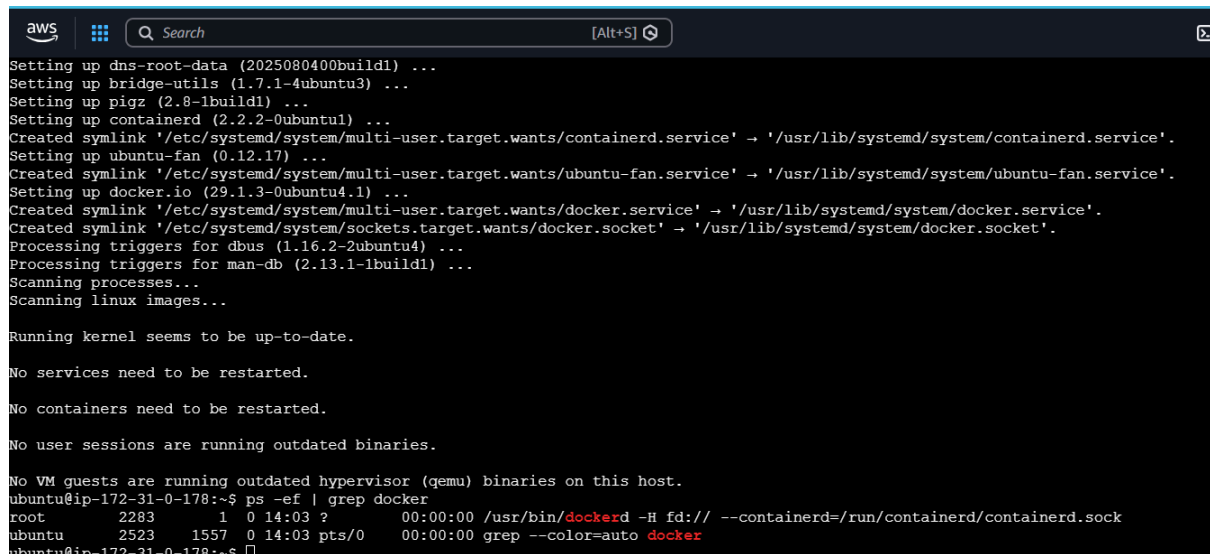
Installing dependencies:
  bridge-utils containerd dns-root-data dnsmasq-base pigz runc ubuntu-fan

Suggested packages:
  ifupdown aufs-tools cgroupfs-mount | cgroup-lite debootstrap docker-buildx docker-compose-v2 docker-doc rinse rootlesskit zfs-fuse | zfsutils

Summary:
  Upgrading: 0, Installing: 8, Removing: 0, Not Upgrading: 21
  Download size: 74.0 MB
  Space needed: 281 MB / 4696 MB available

Continue? [Y/n] Y
Get:1 http://ap-south-1-ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 pigz amd64 2.8-1build1 [68.5 kB]
Get:2 http://ap-south-1-ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 bridge-utils amd64 1.7.1-4ubuntu3 [34.8 kB]
Get:3 http://ap-south-1-ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 runc amd64 1.4.0-0ubuntu1 [9829 kB]
Get:4 http://ap-south-1-ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 containerd amd64 2.2.2-0ubuntu1 [28.0 MB]
Get:5 http://ap-south-1-ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 dns-root-data all 2025080400build1 [6022 B]
Get:6 http://ap-south-1-ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 dnsmasq-base amd64 2.92-1 [441 kB]
Get:7 http://ap-south-1-ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64v3 docker.io amd64 29.1.3-0ubuntu4.1 [35.5 MB]
Get:8 http://ap-south-1-ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 ubuntu-fan all 0.12.17 [34.3 kB]
Fetched 74.0 MB in 1s (73.0 MB/s)
Preconfiguring packages ...
Selecting previously unselected package pigz.
(Reading database ... 85301 files and directories currently installed.)
```

Docker daemon is running



```
Setting up dns-root-data (2025080400build1) ...
Setting up bridge-utils (1.7.1-4ubuntu3) ...
Setting up pigz (2.8-1build1) ...
Setting up containerd (2.2.2-0ubuntu1) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/containerd.service' -> '/usr/lib/systemd/system/containerd.service'.
Setting up ubuntu-fan (0.12.17) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/ubuntu-fan.service' -> '/usr/lib/systemd/system/ubuntu-fan.service'.
Setting up docker.io (29.1.3-0ubuntu4.1) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/docker.service' -> '/usr/lib/systemd/system/docker.service'.
Created symlink '/etc/systemd/system/sockets.target.wants/docker.socket' -> '/usr/lib/systemd/system/docker.socket'.
Processing triggers for dbus (1.16.2-2ubuntu4) ...
Processing triggers for man-db (2.13.1-1build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-0-178:~$ ps -ef | grep docker
root      2283      1   0 14:03 ?        00:00:00 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock
ubuntu    2523    1557   0 14:03 pts/0    00:00:00 grep --color=auto docker
ubuntu@ip-172-31-0-178:~$
```

```
ubuntu@ip-172-31-0-178:~$ ps -ef | grep docker
root      2283      1   0 14:03 ?        00:00:00 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock
ubuntu    2523    1557   0 14:03 pts/0    00:00:00 grep --color=auto docker
ubuntu@ip-172-31-0-178:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
ubuntu@ip-172-31-0-178:~$
```

Validated docker command and made sure it is working fine

```
aws | Search [Alt+S]
root@ip-172-31-0-178:~# docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
root@ip-172-31-0-178:~# docker images
IMAGE ID          DISK USAGE  CONTENT SIZE  EXTRA
root@ip-172-31-0-178:~#
root@ip-172-31-0-178:~# docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:f9078146db2e05e794366b1bfe584a14ea6317f44027d10ef7dad65279026885
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.
```

Checked status and logs for existing container

```
root@ip-172-31-0-178:~# docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
f40d67637969   hello-world   "/hello"   8 minutes ago   Exited (0) 8 minutes ago   vibrant_meitner

root@ip-172-31-0-178:~# docker logs f40d67637969

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/
```

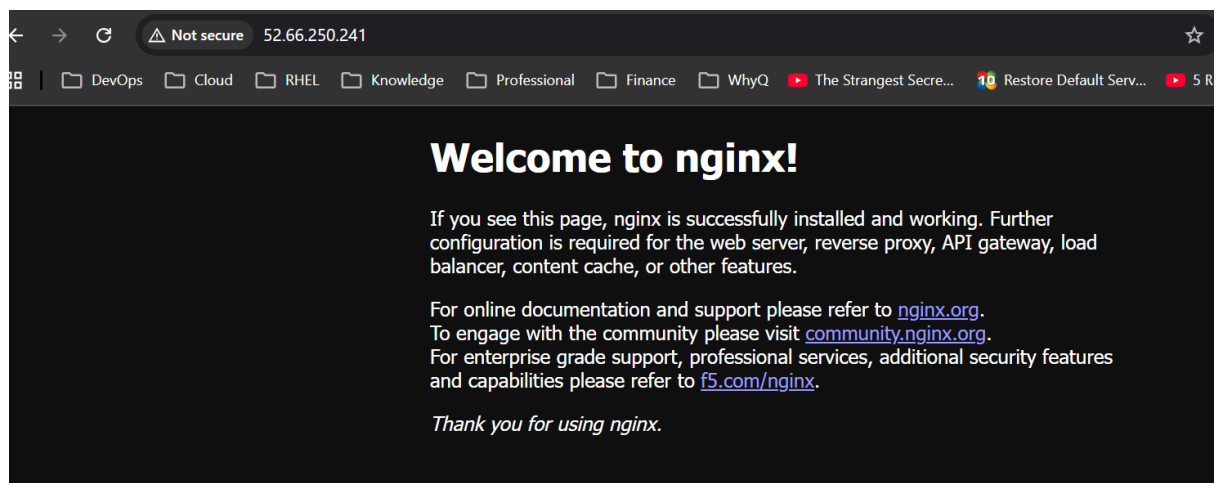
Searched for docker ubuntu image

```
root@ip-172-31-0-178:~# docker search ubuntu
NAME                DESCRIPTION                STARS   OFFICIAL
ubuntu              Ubuntu is a Debian-based Linux operating sys... 17820   [OK]
ubuntu/squid        Squid is a caching proxy for the Web. Long-t... 127
ubuntu/nginx        Nginx, a high-performance reverse proxy & we... 141
ubuntu/kafka        Apache Kafka, a distributed event streaming ... 60
ubuntu/bind9        BIND 9 is a very flexible, full-featured DNS... 118
ubuntu/apache2      Apache, a secure & extensible open-source HT... 103
ubuntu/prometheus   Prometheus is a systems and service monitori... 80
ubuntu/zookeeper    ZooKeeper maintains configuration informatio... 14
ubuntu/jre          Chiselled Java runtime based on Ubuntu. Long... 25
ubuntu/dotnet-aspnet Chiselled Ubuntu runtime image for ASP.NET a... 26
ubuntu/postgres     PostgreSQL is an open source object-relatio... 44
ubuntu/mysql        MySQL open source fast, stable, multi-thread... 74
ubuntu/python       A chiselled Ubuntu rock with the Python runt... 33
ubuntu/redis        Redis, an open source key-value store. Long... 25
ubuntu/dotnet-runtime Chiselled Ubuntu runtime image for .NET apps... 24
ubuntu/dotnet-deps   Chiselled Ubuntu for self-contained .NET & A... 16
ubuntu/grafana       Grafana, a feature rich metrics dashboard & ... 12
ubuntu/memcached     Memcached, in-memory keyvalue store for smal... 7
ubuntu/jdk          Ubuntu rock for OpenJDK 0
ubuntu/mimir       Ubuntu ROCK for Mimir, a horizontally scalab... 0
ubuntu/loki         Grafana Loki, a log aggregation system like ... 2
ubuntu/rabbitmq     Ubuntu-based RabbitMQ image - an open source... 0
ubuntu/grafana-agent Ubuntu ROCK for Grafana Agent, an open-sourc... 0
ubuntu/alertmanager Ubuntu Rock for Alertmanager. It handles ale... 1
ubuntu/opentelemetry-collector This is a rock for the OpenTelemetry Collect... 0
```

Running new nginx image in detached mode and checking status

```
root@ip-172-31-0-178:~# docker run -d -p 80:80 --name=kd-nginx nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
801a1ad15b4e: Pull complete
3531af2bc2a9: Pull complete
ce776bbcdad0d: Pull complete
85c66128325a: Pull complete
4677c2a9a3d4: Pull complete
ff048f1f2159: Pull complete
677c63196868: Pull complete
54c38c75806e: Download complete
69989ccd189b: Download complete
Digest: sha256:6e23479198b998e5e25921dff8455837c7636a67111a04a635cf1bb363d199dc
Status: Downloaded newer image for nginx:latest
2eac83b2bdd714b9c21d6bc0812a2b59ed884242de5ac45844234a0d24c07cd3
root@ip-172-31-0-178:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
2eac83b2bdd7   nginx     "/docker-entrypoint. ..." 13 seconds ago Up 12 seconds 0.0.0.0:80->80/tcp, [::]:80->80/tcp kd-nginx
root@ip-172-31-0-178:~#
```

Validated nginx running on my ecc2 instance



Stopped and started the nginx docker container.

```
root@ip-172-31-0-178:~/snap/amazon-ssm-agent/common# docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
2eac83b2bdd7   nginx     "/docker-entrypoint. ..." 11 minutes ago Up 11 minutes 0.0.0.0:80->80/tcp, [::]:80->80/tcp kd-nginx
f40d67637969   hello-world "/hello"               45 minutes ago Exited (0) 45 minutes ago          vibrant_meitner
root@ip-172-31-0-178:~/snap/amazon-ssm-agent/common# docker stop 2eac83b2bdd7
2eac83b2bdd7
root@ip-172-31-0-178:~/snap/amazon-ssm-agent/common#
root@ip-172-31-0-178:~/snap/amazon-ssm-agent/common#
root@ip-172-31-0-178:~/snap/amazon-ssm-agent/common# docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
2eac83b2bdd7   nginx     "/docker-entrypoint. ..." 12 minutes ago Exited (0) 6 seconds ago          kd-nginx
f40d67637969   hello-world "/hello"               46 minutes ago Exited (0) 46 minutes ago          vibrant_meitner
root@ip-172-31-0-178:~/snap/amazon-ssm-agent/common# docker start 2eac83b2bdd7
2eac83b2bdd7
root@ip-172-31-0-178:~/snap/amazon-ssm-agent/common#
root@ip-172-31-0-178:~/snap/amazon-ssm-agent/common#
root@ip-172-31-0-178:~/snap/amazon-ssm-agent/common# docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
2eac83b2bdd7   nginx     "/docker-entrypoint. ..." 12 minutes ago Up 4 seconds 0.0.0.0:80->80/tcp, [::]:80->80/tcp kd-nginx
f40d67637969   hello-world "/hello"               46 minutes ago Exited (0) 46 minutes ago          vibrant_meitner
root@ip-172-31-0-178:~/snap/amazon-ssm-agent/common#
```

Interacting with the a running container

```
see 'docker exec --help' for more information
root@ip-172-31-0-178:~/snap/amazon-ssm-agent/common# docker exec -it 2eac83b2bdd7 bash
root@2eac83b2bdd7:/# ls
bin boot dev docker-entrypoint.d docker-entrypoint.sh etc home lib lib64 media mnt opt proc root run sbin srv sys tmp usr var
root@2eac83b2bdd7:/# ls -ltr
total 64
lrwxrwxrwx 1 root root 8 Mar 2 21:50 sbin -> usr/sbin
lrwxrwxrwx 1 root root 9 Mar 2 21:50 lib64 -> usr/lib64
lrwxrwxrwx 1 root root 7 Mar 2 21:50 lib -> usr/lib
drwxr-xr-x 2 root root 4096 Mar 2 21:50 home
drwxr-xr-x 2 root root 4096 Mar 2 21:50 boot
lrwxrwxrwx 1 root root 7 Mar 2 21:50 bin -> usr/bin
drwxr-xr-x 1 root root 4096 Apr 21 00:00 var
drwxr-xr-x 1 root root 4096 Apr 21 00:00 usr
drwxrwxrwt 2 root root 4096 Apr 21 00:00 tmp
drwxr-xr-x 2 root root 4096 Apr 21 00:00 srv
drwx----- 2 root root 4096 Apr 21 00:00 root
```

```

root@2eac83b2bdd7:/usr/share/nginx/html# ls -lrt
total 12
-rw-r--r-- 1 root root 896 Apr  7 11:37 index.html_orig
-rw-r--r-- 1 root root 896 Apr  7 11:37 index.html
-rw-r--r-- 1 root root 497 Apr  7 11:37 50x.html
root@2eac83b2bdd7:/usr/share/nginx/html# pwd
/usr/share/nginx/html
root@2eac83b2bdd7:/usr/share/nginx/html# 

```

Checking docker info:

```

root@ip-172-31-0-178:~/snap/amazon-ssm-agent/common# docker info
Client:
 Version:      29.1.3
 Context:      default
 Debug Mode:   false
 Plugins:
  trust: Manage trust on Docker images (Docker Inc.)
        Version:  29.1.3
        Path:     /usr/libexec/docker/cli-plugins/docker-trust

Server:
 Containers: 2
  Running: 1
  Paused: 0
  Stopped: 1
 Images: 2
 Server Version: 29.1.3
 Storage Driver: overlayfs
  driver-type: io.containerd.snapshotter.v1
 Logging Driver: json-file
 Cgroup Driver: systemd
 Cgroup Version: 2
 Plugins:
  Volume: local
  Network: bridge host ipvlan macvlan null overlay
  Log: awslogs fluentd gcplogs gelf journald json-file local splunk syslog
 CDI spec directories:
  /etc/cdi

```

Removing docker containers

```

root@ip-172-31-0-178:~# docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS   NAMES
fc96343850ba   nginx    "/docker-entrypoint..." 8 minutes ago Exited (255) 5 minutes ago          kd-nginx-with-volume
90a29f952e8e   nginx    "/docker-entrypoint..." 12 minutes ago Created                               kd-nginx
2eac83b2bdd7   nginx    "/docker-entrypoint..." About an hour ago Exited (0) 15 minutes ago          kd-nginx
f40d67637969   hello-world "/hello"                 2 hours ago    Exited (0) 2 hours ago          vibrant_meitner

root@ip-172-31-0-178:~# docker rm fc96343850ba
fc96343850ba
root@ip-172-31-0-178:~# docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS   NAMES
90a29f952e8e   nginx    "/docker-entrypoint..." 13 minutes ago Created                               kd-nginx
2eac83b2bdd7   nginx    "/docker-entrypoint..." About an hour ago Exited (0) 17 minutes ago          kd-nginx
f40d67637969   hello-world "/hello"                 2 hours ago    Exited (0) 2 hours ago          vibrant_meitner
root@ip-172-31-0-178:~#

```

Inspecting docker network settings and volume:

```
aws [Search] [Alt+S] Ask Amazon Q
},
"NetworkSettings": {
  "SandboxID": "c03d65b3081c0af91231339df8e8f27b34e1d7072411697f727a4e16273462a0",
  "SandboxKey": "/var/run/docker/netns/c03d65b3081c",
  "Ports": {
    "80/tcp": [
      {
        "HostIp": "0.0.0.0",
        "HostPort": "80"
      },
      {
        "HostIp": "::",
        "HostPort": "80"
      }
    ]
  },
  "Networks": {
    "bridge": {
      "IPAMConfig": null,
      "Links": null,
      "Aliases": null,
      "DriverOpts": null,
      "GwPriority": 0,
      "NetworkID": "7dde78de407cab09c2042bbd3f401b8a39e81ade5a61f524d4dde1aaf6b61ea",
      "EndpointID": "04a5caf66b79c6adfcddcf2badb0a13ea9df0fcd1cfda0dec8a56414c683f84d",
      "Gateway": "172.17.0.1",
      "IPAddress": "172.17.0.2",
      "MacAddress": "6a:70:33:9f:19:81",
    }
  }
}
```

Fetching container ip

```
root@ip-172-31-0-178:~# docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
90a29f952e8e   nginx    "/docker-entrypoint..." 21 minutes ago Created                               kd-nginx
2eac83b2bdd7   nginx    "/docker-entrypoint..." About an hour ago Up 5 minutes    0.0.0.0:80->80/tcp, [::]:80->80/tcp kd-nginx
f40d67637969   hello-world "/hello"                2 hours ago   Exited (0) 2 hours ago                               vibrant_meitner
root@ip-172-31-0-178:~# docker inspect -f '{{range .NetworkSettings.Networks}}{{.IPAddress}}{{end}}' 2eac83b2bdd7
172.17.0.2
root@ip-172-31-0-178:~#
```